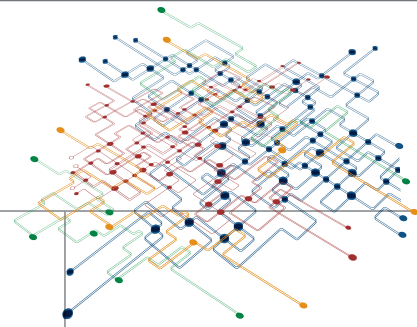


AlliedWare Plus

Layer 3 Operating System (running on switches)



AlliedWare Plus is the next generation operating system from Allied Telesis, providing advanced IPv4 and IPv6 features, superior robustness, and ease of management. AlliedWare Plus also runs on our Firewalls and Wireless APs (which are not covered in this datasheet).

Overview

AlliedWare Plus is an industry-leading operating system that delivers more features and robustness by employing a modular approach to software creation. This approach enables Allied Telesis to implement new functionality faster with exceptional quality, which means customers can upgrade their networks with minimal disruption.

The AlliedWare Plus operating system combines superior networking functionality and strong management capabilities with the exceptional performance that today's networks demand. A standards-based implementation, it also assures full interoperability with other major network equipment, and features improved usability for a superior customer experience.

Modularity + monitoring = robust flexibility

AlliedWare Plus features a modular architecture, providing superior reliability. This architecture uses separate software processes, called modules, to handle different functions such as management, routing protocols, and control. Each of these modules can access only its allocated memory, preventing processes from corrupting each other and causing system crashes. Although independent of each other, modules communicate via well-defined interfaces.

Unified network management

AlliedWare Plus can manage large-scale wired and wireless networks on a single platform to reduce complexity and increase administrative consistency. The Allied Telesis Autonomous Management Framework™ Plus (AMF Plus) is the key to unifying network management. It saves time and reduces cost by automating many every day network management tasks.

Management of Allied Telesis TQ Series wireless access points is now possible directly from AlliedWare Plus switches with the Wireless Manager. Provisioning, operation, administration, and maintenance for the entire enterprise wireless infrastructure, can be performed centrally thereby reducing TCO and improving the user experience.

Certified secure

Allied Ware Plus uses strong cryptographic algorithms with large key sizes to support secure communication and for protection of information. The cryptographic algorithms used in Allied Ware Plus are CAVP* certified.

Easy to use

The AlliedWare Plus operating system incorporates an industry standard Command Line Interface (CLI), facilitating intuitive manageability. Each command is associated with a specific function or task. Many of the commands can be used in scripts, allowing the automation of configuration tasks. Users can also utilize Triggers, which provide a powerful mechanism for automatic and timed management by automating the execution of commands in response to specific events.

The built-in, web-based Graphical User Interface (GUI) is an easy-to-use and powerful management tool. With comprehensive monitoring facilities and the ability to view a virtual chassis as a single entity, the GUI is an essential part of a network management toolkit.

Key Features

- AMF Plus
- AMF Plus secure mode
- Active Fiber Monitoring (AFM)
- VLAN Mirroring (RSPAN)
- VLAN ACLs
- G.8032 Ethernet Ring Protection
- Media Redundancy Protocol (MRP)
- Ethernet Connectivity Fault Monitoring (CFM)
- Upstream Forwarding Only (UFO)
- Continuous PoE
- Precision Time Protocol (PTP) Transparent Mode
- AWC Wireless Management
- Media Access Control Security (MACSec)
- Modbus support
- Bidirectional Forwarding Detection (BFD)
- Multicast Source Discovery Protocol (MSDP)
- Link Monitoring
- VXLAN static tunnels
- NETCONF/RESTCONF with YANG data modelling
- IPFIX (IP Flow Information Export)
- Data Center Bridging Capabilities Exchange Protocol (DCBX)
- Enhanced Transmission Selection (ETS)
- Priority-based Flow Control (PFC)

Futureproof

AlliedWare Plus meets the needs of government and enterprise customers seeking to ensure their network investment is protected now and into the future. With a forward thinking product roadmap, a full IPv6 feature set and the imminent release of OpenFlow v1.3, long-term network reliability and investment security is guaranteed with Allied Telesis.

* Cryptographic Algorithm Validation Program (CAVP) is administered by the National Institute of Standards and Technology (NIST) in the United States.

KEY FEATURES

Allied Telesis Autonomous Management Framework™ Plus (AMF Plus)

AMF Plus is a sophisticated suite of management tools that provide a simplified approach to network management. Powerful features like centralized management, auto-backup, auto-upgrade, auto-provisioning and auto-recovery enable plug-and-play networking and zero-touch management.

The AMF Plus master enables auto-provisioning and auto-upgrade by providing appropriate files to new network members. New network devices can be pre-provisioned making installation easy because no on-site configuration is required.

AMF Plus secure mode encrypts all AMF Plus traffic, provides unit and user authorization, and monitors network access to greatly enhance network security.

An AMF Plus license (from AW+ version 5.5.2-2 onwards) provides all standard AMF network management and automation features, and also enables the AMF Plus intent-based networking features in Vista Manager EX (from version 3.10.1 onwards).

AMF Plus Controller

An AMF Plus master can manage networks of up to 300 nodes, which can be located locally or across WAN links. This can be dramatically increased by installing the AMF Plus Controller, which enables multiple AMF Plus Masters to be managed from a single point. With the AMF Plus Controller, a network of up to 18,000 devices can be managed, allowing all the time saving, cost reducing benefits of AMF Plus to be multiplied and efficiencies to be increased.

AMF Plus Cloud

AMF Plus Cloud allows the AMF Plus Master and/or Controller to be virtual appliances rather than integrated into an Allied Telesis switch or firewall. AMF Plus Cloud offers full AMF Plus functionality, with the advantages of cloud-based access and flexibility.

Multiple Tenants on AMF Plus Cloud

Support multiple tenants (up to 300) on AMF Plus Cloud. Each tenant network is kept separate from other tenant networks for fully flexible deployment, with central or individual network management options. As an AMF Plus Master capable device is not required in every location, AMF Plus multi-tenant provides a high-value solution for large distributed companies, as well as service providers offering network provisioning and/or management services.

Virtual Routing and Forwarding (VRF Lite)

VRF Lite provides Layer 3 network virtualization by dividing a single router into multiple independent virtual routing domains. With independent routing domains, IP addresses can overlap without causing conflict, allowing multiple customers to have their own secure virtual network within the same physical infrastructure. VRF Lite supports IPv4 (unicast and multicast) and IPv6 (unicast) traffic.

The DHCP Server built-in to many products is VRF aware, enabling the supply of IP addresses to clients across multiple isolated networks.

VCStack™ (Virtual Chassis Stacking)

VCStack makes networking simple. It allows multiple switches to be connected together via high speed stacking links. This aggregates the switches, which

then appear as a single switch, or 'virtual chassis'. The virtual chassis can be configured and managed via a single serial console or IP address, which provides greater ease of management in comparison to an arrangement of individually managed switches, and often eliminates the need to configure protocols like VRRP and Spanning Tree.

Long-distance Stacking

Long-distance stacking allows a VCStack to be created over longer distances, perfect for a distributed network environment.

VCStack Plus™

Two SwitchBlade x8100 chassis can be stacked together into a single virtual unit using VCStack Plus. The stacking link uses the 10 Gigabit front panel ports on the CFC960 control cards, which provides a massive 160 Gigabits of stacking bandwidth. VCStack Plus provides a highly available system where network resources and distribution switches are connected across the units for ultimate resiliency. Management is simplified as the two chassis operate as a single virtual unit.

In-Service Software Upgrade (ISSU)

ISSU (also called "hitless firmware upgrade") allows firmware to be updated without causing any network disruption from a device reboot. This enables essential maintenance to be performed when it is required rather than having to schedule a network outage or tolerate any loss of service. ISSU is supported on dual controller systems and can be used in conjunction with VCStack Plus, making it ideal for high availability applications.

Link Aggregation

Link aggregation allows a number of individual switch ports to be combined, forming a single logical connection of higher bandwidth. This provides a higher performance link, and also provides redundancy for a more reliable and robust network.

VCStack and Link Aggregation

Link aggregation can be used across members of a stack to protect against link and device failures. This provides a resilient network solution that is easier to implement and administer than traditional redundant core networks. A VCStack solution in combination with link aggregation also achieves load balancing, as the stacked devices share the network traffic.

MEF Certified

Switching products running AlliedWare Plus have been certified by the Metro Ethernet Forum (MEF) certification program, which tests products for conformance to the strict requirements of Carrier Ethernet. Specifically, these products are certified for compliance to MEF 9 and MEF 14 Ethernet services tests.

sFlow

sFlow is an industry-standard technology for monitoring high-speed switched networks. It provides complete visibility into network use, enabling performance optimization, usage accounting/billing, and defense against security threats. Sampled packets sent to a collector (up to 5 collectors can be configured) ensure it always has a real-time view of network traffic.

IPFIX (IP Flow Information Export)

IPFIX enables exporting IP flow data in a network for analysis. This provides network administrators with information for accounting, billing, capacity planning, and performance optimization.

Bi-directional Forwarding Detection (BFD)

BFD enables fast detection of link failures, so recovery time is minimized. BFD works with static routes, and also alongside BGP and OSPF dynamic routing protocols supporting faster shutdown of neighbor connections if a peer session goes down. When using VRF-Lite, BFD is supported globally or within a domain.

802.1x, RADIUS Authentication and Dynamic VLAN Assignment

The IEEE 802.1x standard manages port-based network access. It provides authentication to devices attached to a LAN port by initiating a connection, or preventing access if authentication fails. Valuable for authenticating and controlling user traffic to a protected network, 802.1x is also effective for dynamically varying encryption keys. 802.1x attaches the Extensible Authentication Protocol (EAP) to both wired and wireless LAN media, and supports multiple authentication methods, such as token cards, Kerberos, certificates, and public key authentication.

802.1x uses the RADIUS (Remote Authentication Dial In User Service) protocol to transfer authentication and configuration information between the switch and a shared RADIUS authentication server, which manages a user database and provides information to the client.

Dynamic VLAN assignment allows an 802.1x supplicant to be placed into a specific VLAN, based on information returned from the RADIUS server during authentication. This limits a supplicant's network access to a specific VLAN, and prevents supplicants from connecting to VLANs for which they are not authorized.

Dynamic ACLs assigned via port authentication.

Access Control Lists (ACLs)

AlliedWare Plus delivers industry-standard access control functionality through Access Control Lists (ACLs). ACLs filter network traffic to control whether packets are forwarded or blocked at the port interface. The switch examines each packet to determine whether to forward or drop the packet, based on criteria that is specified within the ACL, such as source and destination MAC or IP address, IP protocol, or TCP/UDP port. This provides a powerful network security mechanism to select the types of traffic to be analyzed, forwarded, or influenced in some way, for example, to restrict routing updates or provide traffic flow control.

VLAN Access Control List (ACLs)

ACLs simplify access and traffic control across entire segments of the network. They can be applied to a VLAN as well as a specific port.

Bridge Protocol Data Unit (BPDU) Protection

BPDU protection adds extra security to the Spanning Tree Protocol (STP). It protects the spanning tree configuration by preventing malicious DoS attacks caused by spoofed BPDUs.

BPDU Protection is designed to be enabled on ports that should not receive BPDUs. These are edge ports connected to end user devices that do not run

spanning tree. If a spoofed BPDU packet is received on a protected port, the BPDU Protection feature disables the port and alerts the network manager.

EPSRings™ (Ethernet Protection Switched Ring)

EPSRing and 10 Gigabit Ethernet allow several switches to form high-speed protected rings, capable of recovery within as little as 50ms. This feature is perfect for high performance and high availability in enterprise networks.

SuperLoop Prevention (SLP) enables a link between two EPSR nodes to be in separate EPSR domains, improving redundancy and network fault resiliency.

G.8032 Ethernet Ring Protection

G.8032 provides standards-based high-speed ring protection, that can be deployed stand-alone, or interoperate with Allied Telesis EPSR.

Ethernet Connectivity Fault Monitoring (CFM) proactively monitors links and VLANs, and provides alerts when a fault is detected.

Storm Protection

Bandwidth limiting minimizes the effects of the storm by reducing the amount of flooding traffic.

Policy-based storm protection is more powerful than bandwidth limiting. It restricts storm damage to within the storming VLAN, and allows the traffic rate that creates a broadcast storm to be defined. The action the device should take when it detects a storm, such as disabling the port from the VLAN or shutting the port down, can also be configured.

Packet storm protection allows limits to be set on the broadcast reception rate, multicast frames and destination lookup failures. In addition, separate limits can be specified when the device will discard each of the different packet types.

Loop Protection

Thrash Limiting, also known as rapid MAC movement, detects and resolves network loops. It is highly user-configurable — from the rate of looping traffic to the type of action the switch should take when it detects a loop.

Loop Detection works in conjunction with Thrash Limiting. With Thrash Limiting, the switch only detects a loop when a storm has occurred, which can potentially cause disruption to the network. To avoid this, Loop Detection sends special Loop Detection Frame (LDF) packets that the switch listens for. If an LDF packet is received, then the switch can be configured to either disable the port, disable the link, or send an SNMP trap.

Policy-Based Quality of Service (QoS)

Comprehensive, low latency QoS features operating at wire-speed provide flow-based traffic management with full classification, prioritization, traffic shaping and min/max bandwidth profiles.

Our QoS features are ideal for Service Providers wanting to ensure maximum availability of premium voice, video and data services, and at the same time manage customer Service Level Agreements (SLAs).

For enterprise customers, QoS features protect productivity by guaranteeing performance of business-critical applications (including VoIP services), and help to restore and maintain the responsiveness of enterprise applications in the workplace.

Link Layer Discovery Protocol - Media Endpoint Discovery (LLDP - MED)

LLDP-MED extends LLDP's basic network endpoint discovery and management functions. LLDP-MED allows for media end-point specific messages, providing detailed information on power requirement, network policy, location discovery (for emergency call services) and inventory. LLDP-MED is an important feature for simplifying VoIP, security camera and WLAN deployments.

Power over Ethernet Plus (PoE+)

With PoE, a separate power connection to end points such as IP phones and wireless access points is not necessary. PoE+ provides up to 30 Watts per port, while PoE++ provides up to 90 Watts per port for even greater flexibility, and the ability to connect and power devices such as lighting controllers and ultra high resolution PTZ cameras.

Dynamic Host Configuration Protocol (DHCPv6)

DHCPv6 is used to dynamically assign IPv6 addresses to hosts from a central location. Acting as DHCPv6 client enables the switch to receive an IPv6 address, and acting as server enables the switch to dynamically allocate IPv6 addresses to hosts. The DHCPv6 server and client both support the Prefix Delegation feature, which allocates a whole IPv6 subnet to a DHCP client. The client, in turn, can allocate addresses from this subnet to the hosts that are connected to it.

Find Me

In busy server rooms consisting of a large number of equipment racks, it can be quite a job finding the correct switch quickly among many similar units. The "find me" feature is a simple visual way to quickly identify the desired physical switch for maintenance or other purposes, by causing its LEDs to flash in a specified pattern.

Active Fiber Monitoring

Active Fiber Monitoring detects tampering on optical links by regularly checking received optical power. This provides increased security by ensuring the integrity of short- and long-haul fiber links.

VLAN Mirroring (RSPAN)

VLAN mirroring allows traffic from a port on a remote switch to be analysed locally. Traffic being transmitted or received on the port is duplicated and sent across the network on a special VLAN.

Upstream Forwarding Only (UFO)

UFO lets you manage which ports in a VLAN can communicate with each other, and which only have upstream access to services, for secure multi-user deployment.

Continuous PoE

Continuous PoE allows the switch to be restarted without affecting the supply of power to connected devices. Smart lighting, security cameras, and other PoE devices will continue to operate during a software upgrade on the switch.

Precision Time Protocol (PTP)

PTP (IEEE 1588) synchronizes clocks throughout the network with micro-second accuracy, supporting industrial automation and control systems. PTP operates on standalone or stacked switches.

Wireless Management

Autonomous Wave Control (AWC) wireless controller manages a wireless network, and automatically optimizes AP performance.

Media Access Control Security (MACSec)

802.1AE MACSec secures all traffic on point-to-point Ethernet links between directly connected nodes, ensuring protection against security threats such as denial of service, intrusion, man-in-the-middle, passive wiretapping, and playback attacks.

Modbus

Modbus enables communication with Supervisory Control and Data Acquisition (SCADA) systems for industrial automation.

Media Recovery Protocol (MRP)

MRP enables high-availability automation networks, and is specified for rings with up to 50 devices, where it guarantees fully deterministic switchover behavior.

Multicast Source Discovery Protocol (MSDP)

MSDP enables two or more PIM-SM (Sparse Mode) domains to share information on active multicast sources, for more efficient forwarding of multicast traffic.

Link Monitoring (Linkmon)

Linkmon enables network health monitoring by regularly sending probes over key links to gather metrics comprising latency, jitter, and probe loss. This supports pro-active network management, and can also be used with triggers to automate a change to device or network configuration in response to the declining health of a monitored link.

Virtual Extensible LAN (VXLAN)

tunnels

VXLAN tunnels let you join two or more L2 networks over an L3 IP network to form a single L2 broadcast domain. VXLAN adds scalability to cloud computing environments. Static VXLAN tunnels are supported on the SBx908GEN2, x950 and x530 Series switches.

NETCONF/RESTCONF

NETCONF/RESTCONF with YANG data modeling provide a standardized way to represent data, and securely configure devices from modern management systems..

Data center Quality of Service

Data Center Bridging Capabilities Exchange Protocol (DCBX) lets network devices in the data center automatically discover and negotiate settings for DCB features, ensuring optimized performance.

Enhanced Transmission Selection (ETS) provides quality of service by allocating bandwidth to important traffic classes, with the flexibility to share remaining bandwidth to maximize traffic throughput and performance.

Priority-based Flow Control (PFC) enables lossless data forwarding by managing high priority traffic queues to support their class-of-service.

Feature Licenses for switches running AlliedWare Plus 5.5.6-1

	Premium License	Standalone Licenses	
CentreCOM GS970EMX Series	AT-FL-G97EMX-01 Static Route (128 routes) RIP (256 Routes) OSPFv2 (128 routes) PIMv4-SM, DM and SSM EPSR master		
CentreCOM GS970M Series		AT-FL-GS97-UD UniDirectional Link Detection	
CentreCOM GS980EM Series	AT-FL-G98EM-01 OSPF (128 Routes) PIMv4-SM, DM and SSM RIP (256 Routes) STATIC (128 Routes) EPSR master		
CentreCOM GS980M Series		AT-FL-GS98M-CP Continuous PoE	
CentreCOM GS980MX Series		AT-FL-G98MX-CP Continuous PoE AT-FL-G98MX-UD UniDirectional Link Detection	
CentreCOM XS900MX Series	AT-FL-XS9MX-01 IPv4 Static routing (128 routes) RIP (256 routes) OSPFv2 (256 routes) PIMv4-SM, DM and SSM EPSR master	AT-FL-XS9X-UDLD UniDirectional Link Detection	
x220 Series		AT-FL-x220-UDLD UniDirectional Link Detection AT-FL-x220-8032 G.8032 ring protection	AT-FL-x220-CPOE Continuous PoE for 52GP model
x240 Series		AT-FL-x240-APP VLAN double tagging (QinQ) PTP transparent mode AT-FL-x240-OF13¹ OpenFlow v1.3	
x250 Series		AT-FL-x250-APP VLAN double tagging (QinQ) PTP transparent mode VLAN translation	AT-FL-x250-OF13¹ OpenFlow v1.3
x320 Series	AT-FL-x320-01 OSPF (256 routes) PIMv4-SM, DM and SSM RIPng (256 routes) OSPFv3 (256routes) PIMv6-SM and SSM MLD v1/v2 VLAN double tagging (Q-in-Q)	AT-FL-x320-8032 G.8032 ring protection Ethernet CFM AT-FL-x320-CPOE Continuous PoE	
x330 Series	AT-FL-x330-01 OSPF (256 routes) PIM-DM/SM/SSM (1,023 Entries) RIPng (256 routes) OSPFv3 (256routes) PIM-SM/SSM (1,023 Entries) MLD v1/v2 VLAN double tagging (Q-in-Q) VLAN translation	AT-FL-x330-8032 G.8032 ring protection Ethernet CFM AT-FL-x330-OF13¹ OpenFlow v1.3	

¹ Subscription licenses are available for 1 year or 5 years

Feature Licenses for switches running AlliedWare Plus 5.5.6-1

	Premium License	Standalone Licenses	
x530L Series	AT-FL-x530L-01 OSPFv2 (256 routes) BGP4 (256 routes) PIMv4-SM, DM and SSM v4 VLAN double tagging (Q-in-Q) RIPng (256 routes) OSPFv3 (256 routes) MLDv1/v2 PIM-SMv6/SSMv6 RADIUS-Full UDLD VLAN Translation PTP Transparent mode	AT-SW-APM10^{1, 2} AMF Plus Master (up to 10 nodes) AT-FL-x530L-OF13¹ OpenFlow v1.3 AT-FL-x53L-8032 G.8032 ring protection Ethernet CFM AT-FL-x53L-CPOE Continuous PoE AT-FL-x53L-MSTK Mixed Stacking	
x530 Series	AT-FL-x530-01 OSPFv2 (12,000 routes) BGP4/4+ (5,000 routes) PIMv4-SM, DM and SSM v4 VLAN double tagging (Q-in-Q) RIPng (5,000 routes) OSPFv3 (6,000 routes) MLDv1/v2 PIM-SMv6/SSMv6 RADIUS-Full VRF-Lite (64 domains) UDLD VLAN Translation VXLAN	AT-SW-APM10^{1, 2} AMF Plus Master (up to 10 nodes) AT-FL-x530-OF13¹ OpenFlow v1.3 AT-FL-x530-8032 G.8032 ring protection Ethernet CFM AT-FL-x530-CPOE Continuous PoE for GPXm model AT-FL-x530-MSTK Mixed Stacking	AT-SW-AWC10¹ AWC (up to 10 access points) AT-SW-CB10-2022¹ AWC-CB and AWC-SC (up to 10 access points)
x540L Series	AT-FL-x540L-01 OSPFv2 (256 routes) BGP4 (256 routes) PIMv4-SM, DM and SSM v4 RIPng (256 routes) OSPFv3 (256 routes) MLDv1/v2 PIM-SMv6/SSMv6 VLAN double tagging (Q-in-Q) VLAN Translation PTP Transparent Mode4 RADIUS Full VRF-Lite (64 domains)	AT-SW-APM10^{1, 2} AMF Plus Master (up to 10 nodes) AT-SW-AWC10¹ AWC (up to 10 access points) AT-SW-CB10-2022¹ AWC-CB and AWC-SC (up to 10 access points) AT-FL-x540L-OF13¹ OpenFlow v1.3	
x550 Series	AT-FL-x550-01 RIP (256 routes) OSPF (256 routes) BGP (256 routes) PIMv4-SM, DM and SSM EPSR master VLAN double tagging (Q-in-Q) RIPng (256 routes) OSPFv3 (256 routes) MLDv1 and v2 PIMv6-SM UDLD PTP Transparent mode	AT-SW-APM10^{1, 2} AMF Plus Master (up to 10 nodes) AT-FL-x550-OF13¹ OpenFlow v1.3 AT-FL-x550-8032 G.8032 ring protection Ethernet CFM AT-FL-x550-CPOE Continuous PoE for XSPQ model	AT-SW-AWC10¹ AWC (up to 10 access points) AT-SW-CB10-2022¹ AWC-CB and AWC-SC (up to 10 access points)
x560 Series	AT-FL-x560-01 OSPF (12,000 routes) ³ BGP4 (5,000 routes) ³ PIMv4-SM, DM, and SSM v4 RIPng (12,000 routes) OSPFv3 (12,000 routes) MLDv1/v2 PIM-SMv6/SSMv6 VLAN double tagging (Q-in-Q) VLAN Translation RADIUS Full VRF-Lite (64 domains)	AT-SW-APM10^{1, 2} AMF Plus Master (up to 10 nodes) AT-FL-x560-AAP¹ AMF Application Proxy AT-FL-x560-OF13¹ OpenFlow v1.3	AT-SW-AWC10¹ AWC (up to 10 access points) AT-FL-x560-MSEC Media Access Control Security

¹ Subscription licenses are available for 1 year or 5 years

² From AW+ version 5.5.2-2 onwards, AMF Plus licenses provide all standard AMF network management and automation features. They also enable the AMF Plus intent-based networking features in Vista Manager EX (from version 3.10 onwards)

³ 64 routes are included for free in the base firmware

Feature Licenses for switches running AlliedWare Plus 5.5.6-1

	Premium License	Standalone Licenses	
x930 Series	AT-FL-x930-01 OSPF (16,000 routes) ³ BGP4 (5,000 routes) ³ PIMv4-SM, DM, SSM (2,000 entries) VLAN Q-in-Q RIPng (5,000 routes) OSPFv3 (8,000 routes) BGP4+ for IPv6 (5,000 routes) MLDv1 and v2 UDLD PIMv6-SM, SSM (1,000 entries) VRF lite (64 domains) RADIUS Full ⁴ PTP Transparent mode	AT-SW-APM10^{1, 2} AMF Plus Master (up to 10 nodes) AT-FL-x930-AAP¹ AMF Application Proxy AT-FL-x930-OF13¹ OpenFlow v1.3 AT-FL-x930-8032 G.8032 ring protection Ethernet CFM AT-FL-x930-CPOE Continuous PoE	AT-FL-x930-MODB Modbus for industrial applications AT-FL-x930-MRP Media Redundancy Protocol AT-FL-x930-MSEC Media Access Control Security AT-SW-AWC10¹ AWC (up to 10 access points) AT-SW-CB10-2022¹ AWC-CB and AWC-SC (up to 10 access points)
x950 Series	AT-FL-x950-01 OSPF (16,000 routes) ³ BGP4 (5,000 routes) ³ PIMv4-SM, DM and SSM (2,000 entries) VLAN double tagging (Q-in-Q) RIPng (5,000 routes) OSPFv3 (8,000 routes) BGP4+ (5,000 routes) MLDv1 and v2 PIMv6-SM and SSM (1,000 entries) VRF lite (63 domains) RADIUS Full ⁴ UDLD VLAN Translation VXLAN PTP Transparent mode	AT-SW-APM10^{1, 2} AMF Plus Master (up to 10 nodes) AT-FL-x950-AAP¹ AMF Application Proxy AT-FL-x950-OF13¹ OpenFlow v1.3 AT-FL-x950-8032 G.8032 ring protection Ethernet CF AT-FL-x950-MODB Modbus for industrial applications AT-FL-x950-MSEC Media Access Control Security AT-FL-x950-VLF VRF Lite (600 domains)	AT-FL-x950-ASEC¹ AMF-Sec AT-SW-AWC10¹ AWC (up to 10 access points) AT-SW-CB10-2022¹ AWC-CB and AWC-SC (up to 10 access points)
x980 Series	AT-FL-x980-01 OSPF (16,000 routes) ³ BGP4 (16,000 routes) ³ PIMv4-SM, DM and SSM (2,000 entries) RIPng (8,000 routes) OSPFv3 (8,000 routes) BGP4+ (8,000 routes) PIMv6-SM and SSM (1,000 entries) MLDv1 and v2 RADIUS Full ⁴ UDLD VLAN double tagging (Q-in-Q) VLAN Translation VXLAN PTP Transparent mode VRF lite (64 domains)	AT-SW-APM10^{1, 2} AMF Plus Master (up to 10 nodes) AT-SW-APC10^{1, 2} AMF Plus Controller (up to 10 areas) AT-FL-GEN2-OF13¹ OpenFlow v1.3 AT-FL-GEN2-VLF VRF Lite (600 domains)	
SwitchBlade x908 GEN2	AT-FL-GEN2-01 OSPF (16,000 routes) ³ BGP4 (5,000 routes) ³ PIMv4-SM, DM and SSM (2,000 entries) VLAN double tagging (Q-in-Q) RIPng (5,000 routes) OSPFv3 (8,000 routes) BGP4+ (5,000 routes) MLDv1 and v2 PIMv6-SM and SSM (1,000 entries) VRF lite (63 domains) RADIUS Full ⁴ UDLD VLAN Translation G.8032 resilient rings Ethernet CFM VXLAN PTP Transparent mode	AT-SW-APM10^{1, 2} AMF Plus Master (up to 10 nodes) AT-SW-APC10^{1, 2} AMF Plus Controller (up to 10 areas) AT-FL-GEN2-OF13¹ OpenFlow v1.3 AT-FL-GEN2-MSEC Media Access Control Security AT-FL-GEN2-VLF VRF Lite (600 domains) AT-FL-GEN2-ASEC¹ AMF-Sec AT-SW-CB10-2022¹ AWC-CB and AWC-SC (up to 10 access points)	

¹ Subscription licenses are available for 1 year or 5 years

² From AW+ version 5.5.2-2 onwards, AMF Plus licenses provide standard AMF management and automation features, and enable the AMF Plus intent-based networking features in Vista Manager EX (from version 3.10 onwards)

³ 64 routes are included for free in the base firmware

⁴ RADIUS-Full increases the local RADIUS database limits to 1,000 NAS devices and 5,000 users

⁵ The number of routes available depends on the silicon profile in use

⁶ Available in a future software release

Feature Licenses for switches running AlliedWare Plus 5.5.6-1

	Premium License	Standalone Licenses	
SwitchBlade x908 GEN3	AT-FL-GEN3-01 OSPF ³ (16,000 or 60,000 routes) ⁵ BGP4 ³ (5,000 routes or 16,000 routes) ⁵ PIMv4-SM, DM and SSM (32,000 entries) ⁵ PIMv4-SSM (8,000 entries) ⁵ VLAN double tagging (Q-in-Q) RIPng (5,000 routes) OSPFv3 (8,000 or 30,000 routes) ⁵ BGP4+ (5,000 routes or 16,000 routes) ⁵ MLDv1 and v2 PIMv6-SM and SSM (7,100 entries) VRF lite (63 domains) RADIUS Full ⁴ UDLD VLAN Translation G.8032 resilient rings Ethernet CFM VXLAN ⁶ PTP Transparent mode	AT-SW-APM10^{1,2} AMF Plus Master (up to 10 nodes) AT-SW-APC10^{1,2} AMF Plus Controller (up to 10 areas) AT-FL-GEN3-OF13^{1,6} OpenFlow v1.3 AT-FL-GEN3-MSEC Media Access Control Security AT-FL-GEN3-VLF VRF Lite (600 domains) AT-SW-AWC10¹ AWC (up to 10 access points) AT-SW-CB10-2022¹ AWC-CB and AWC-SC (up to 10 access points)	
SwitchBlade x8100 Series with CFC960 controller card	AT-FL-CFC960-01 OSPF (5K routes or 10K with XLEM) ³ BGP4 (5K routes or 100K with XLEM) ³ PIMv4-SM, DM, SSM VLAN Q-in-Q RIPng (1K routes or 3.5K with XLEM) UDLD OSPFv3 (1K routes or 5K with XLEM) BGP4+ (1K routes or 50K with XLEM) MLDv1 & v2 PIMv6-SM, SSM RADIUS-Full ⁴ VRF lite (64 domains) VLAN Translation	AT-FL-CF9-VCSP VCStack Plus AT-SW-APM10^{1,2} AMF Plus Master (up to 10 nodes) AT-SW-APC10^{1,2} AMF Plus Controller (up to 10 areas) AT-FL-CF9-8032 G.8032 ring protection Ethernet CFM	

¹ Subscription licenses are available for 1 year or 5 years

² From AW+ version 5.5.2-2 onwards, AMF Plus licenses provide standard AMF management and automation features, and enable the AMF Plus intent-based networking features in Vista Manager EX (from version 3.10 onwards)

³ 64 routes are included for free in the base firmware

⁴ RADIUS-Full increases the local RADIUS database limits to 1,000 NAS devices and 5,000 users

⁵ The number of routes available depends on the silicon profile in use

⁶ Available in a future software release

	Premium License	Standalone Licenses	
IE210L Series	AT-FL-IE2L-L2-1 EPSR Master VLAN translation VLAN double tagging (Q-in-Q) UDLD PTP Transparent Mode	AT-FL-IE2-G8032 ITU-T G.8032 Ethernet CFM AT-FL-IE2-OF13¹ OpenFlow v1.3	
IE220 Series	AT-IE220-FL01 10G uplink ports Hi-PoE sourcing PoE++ sourcing		
IE340 Series	AT-IE340-FL01 BGP (64 routes) BGP+ (64 routes) OSPF (64 routes) OSPFv3 (64 routes) PIM-SM, DM and SSM (256 routes) PIMv6-SM and SSM (256 routes) RIP (64 routes) RIPng (64 routes)		
IE360 Series	AT-IE360-FL01 BGP (64 routes) BGP+ (64 routes) OSPF (256 routes) OSPFv3 (256 routes) PIM-SM, DM and SSM (256 routes) PIMv6-SM and SSM (256 routes) RIP (256 routes) RIPng (256 routes)		
IE560 Series	AT-IE560-FL01 BGP, BGP+ (256 routes) OSPF (256 routes) OSPFv3 (256 routes) PIM-SM, DM and SSM (256 routes) PIMv6-SM and SSM (256 routes) RIP (256 routes) RIPng (256 routes)	AT-SW-APM10^{1,2} AMF Plus Master (up to 10 nodes)	

¹ Subscription licenses are available for 1 year or 5 years

² From AW+ version 5.5.2-2 onwards, AMF Plus licenses provide standard AMF management and automation features, and enable the AMF Plus intent-based networking features in Vista Manager EX (from version 3.10 onwards)

AMF Plus Cloud Master and Controller Licenses

AT-AMFCLOUD-PLUS-BASE-1YR²	1 year AMF Plus Master license for up to 10 nodes
AT-AMFCLOUD-PLUS-BASE-5YR²	5 year AMF Plus Master license for up to 10 nodes
AT-AMFCLOUD-PLUS-EX1-1YR²	1 year AMF Plus Master 1 node add-on license (maximum 350 nodes)
AT-AMFCLOUD-PLUS-EX1-5YR²	5 year AMF Plus Master 1 node add-on license (maximum 350 nodes)
AT-AMFCLOUD-PLUS-EX10-1YR²	1 year AMF Plus Master 10 node add-on license (maximum 350 nodes)
AT-AMFCLOUD-PLUS-EX10-5YR²	5 year AMF Plus Master 10 node add-on license (maximum 350 nodes)
AT-AMFCLOUD-PLUS-CTRL-1YR²	1 year AMF Plus Controller 10 areas base/add-on license (maximum 1000 areas)
AT-AMFCLOUD-PLUS-CTRL-5YR²	5 year AMF Plus Controller 10 areas base/add-on license (maximum 1000 areas)

² From AW+ version 5.5.2-2 onwards, AMF Plus licenses provide all standard AMF network management and automation features. They also enable the AMF Plus intent-based networking features in Vista Manager EX (from version 3.10 onwards)